Critical Design Review
Payload Team
Duke AERO 2023/2024

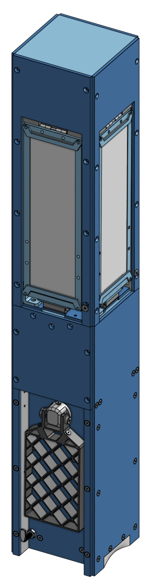
This document is an excerpt from SAC 2024 CDR .

1. General

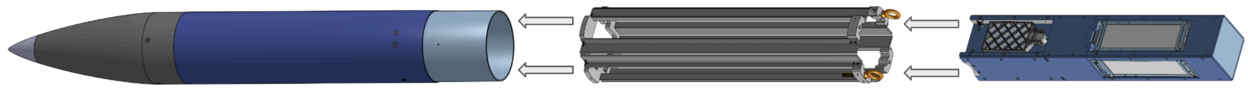
1.1. Overview

The payload is designed for submission to the Space Dynamics Laboratory (SDL) Payload Challenge. The goal of the payload section is to develop an interesting CubeSat and robust deployer system, accommodating many configurations of CubeSats. During flight, a 6U standard CubeSat, “The Log,” will be deployed using this system and begin to perform specified events throughout descent. The deployer will weigh approximately 7.5 lbs and the total CubeSat weight will be approximately 15 lbs, making the entire payload weigh 22.5 lbs. The payload is housed in the forward airframe (nosecone and body tube). It will be approximately 24” long (length of the deployer) and extend from the nosecone, terminating 8” from the end of the aft forward coupler, excluding eyebolts.

The CubeSat model is a standardized form for nano-satellites meant to provide a streamlined platform for satellite-based scientific investigations, experiments, demonstrations, and mission concepts. It is based on the “U” standard, in which 1 “U” corresponds to a 10 cm x 10 cm x 10 cm unit. Scaling up by 10 cm increments in one direction accordingly, CubeSats can be extended up to 12U, and their developments allow for engineers across the world to launch their satellite experiments in an accessible, commercial fashion. Our deployer design is centered around reliability and repeatability, one of the greatest challenges for this type of system. Our 6U CubeSat will showcase multiple technologies such as grid fins for roll control, a variable reefing parachute system (VRS) to control descent velocity throughout flight, and a high-quality live video feed from the CubeSats to monitor vehicle status.



Left: Deployer; Right: The Log



Integration layout

1.2. Design Criteria

The objective of the SDL payload challenge is to “encourage participants to create payloads that accomplish a relevant function and provide useful learning opportunities”. The payload must be a scientific experiment or a technology demonstration. Design constraints are set by challenge rules in addition to mass and size requirements from structures and propulsion teams. The major rules (external criteria) are as follows:

1. The payload must weigh at or greater than 8.8 lbs
2. The payload may not contain more than 2.25 lbs of ballast mass
3. The payload must perform an independent function and may not be a part of launch vehicle performance
4. Payload may be removed and rocket is still operational (i.e., may not include critical avionics)
5. Payloads shall not contain significant quantities of lead or any other hazardous or radioactive materials

2. Structural Integration

2.1. Deployer Integration

Structural rigidity and speed of assembly are paramount given the tight time windows on launch day. The payload must be mounted securely to the forward airframe of the rocket to avoid hindering flight performance or stability in any way.

The entire payload system will be housed in the nosecone and forward airframe of the rocket. The deployer will be fastened with 4, #8-32 screws through the nosecone, forward nosecone joint. The deployer bulkhead will serve as the primary means by which the deployer is fastened to the rest of the rocket. Secondary fasteners will be used at the opposite end of the deployer to keep the deployer rails aligned and support load transfer from 2 eyebolts, which will be loaded during drogue separation.

The deployer will serve only as a means to eject the CubeSats from the rocket. Thereby, the deployer WILL NOT count towards CubeSat mass.

2.2. CubeSat Integration

The CubeSats will be entirely housed in the deployer until drogue separation occurs at apogee. At apogee, The Log will be ejected from the forward airframe, and begin to perform specified functions throughout its descent.

The deployer (more info on mechanism in deployer-specific section below) uses a looped shock-cord system to propel The Log out of the deployer. Before inserting The Log into the deployer, shock cords running from an eyebolt on the recovery bulkhead must be fastened with quick links to the deployer eyebolts on opposite sides of the deployer. Then, both cords will run down the respective side of the deployer, towards the nosecone, and cross paths at the deployer bulkhead. They will then be run up the opposite side of the deployer and back out. For this system to work, at least 1 ft of extra shock cord is required on each side so that once the cords are looped, there is still room to load The Log into the deployer.

3. Deployer

3.1. Principles

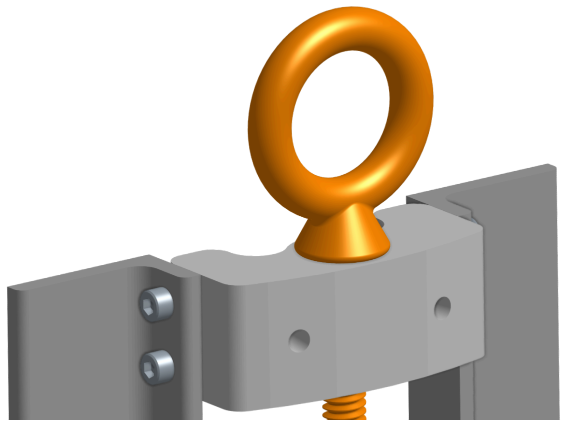
To increase reliability and ease of integration, the deployer was designed to be a purely mechanical system and perform tasks based on redundant signals from other critical rocket events. Taking advantage of the already redundant drogue separation event triggered by avionics, the deployer will eject The Log automatically without relying in any way on its own system for apogee detection and deployment. Using the forces created by drogue separation, The Log will be propelled out of the deployer and forward airframe as recovery shock cords become increasingly taut.

The deployer holds the CubeSats securely in place during launch and once apogee is reached, they will be smoothly deployed. A corner rail system allows for minimal design restrictions on the CubeSats, since they only need to interface with the deployer at the corners.

While it’s very difficult to predict how much pressure and resultant force the ejection charges will produce, we can at least estimate how much snatch force is experienced at drogue deployment. Given some period of time passing between apogee and when drogue deploys and exerts its snatch force, recovery has calculated that for an 80 lb rocket with drogue deployment 3 seconds after apogee, the drogue snatch force would be 31 lbs. With a larger error (5 seconds passing between apogee and deployment) the snatch would be 87 lbs.

3.2. Design

The deployer is designed to be structurally reliant on the forward airframe to keep the “mouth” of the device open. The rail system, which the CubeSats slide on, is fixed to the airframe at the forward and aft of the ends of the deployer. At the nosecone end, 4 #8-32 screws run radially through the nosecone and forward coupler to the bulkhead. This fixture point bears little load since there should be minimal force down the length of the rails. On the aft side of the airframe, each eyebolt is fixed through the lower coupler on a metal bracket using 2 #8-32 bolts. This metal bracket also holds the rails in place.



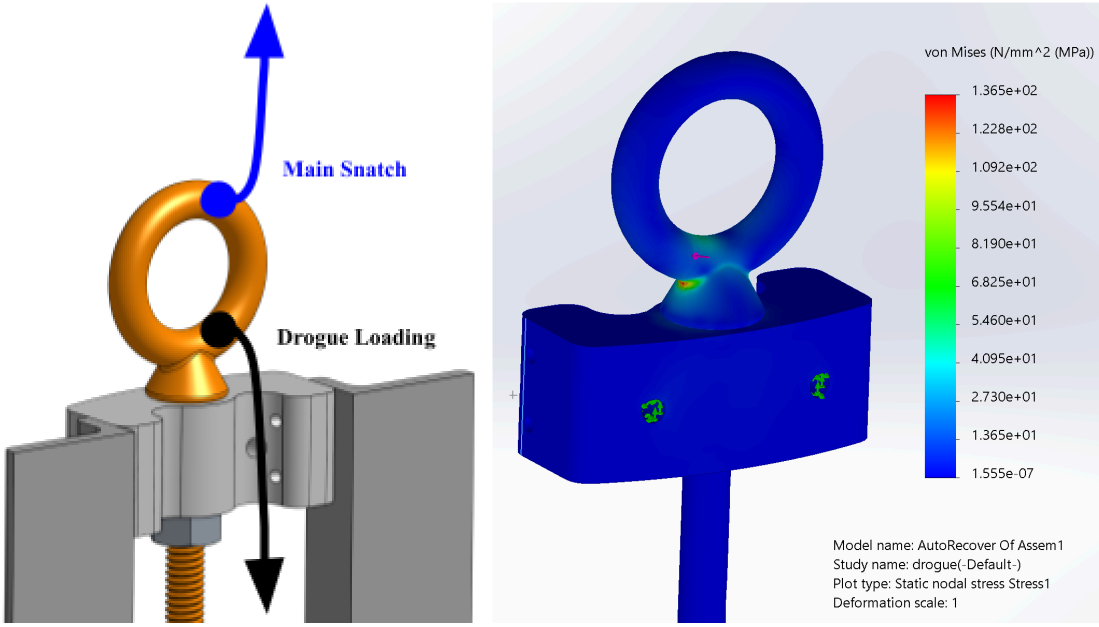
Eyebolt and rail bracket

Along the sides of the forward airframe, 8–24” rails form corners to create a smooth sliding surface for the CubeSats to slide on as they leave the airframe. Testing will be done to ensure smooth sliding of the G10 fiberglass along these 6061 aluminum rails, but initial evaluation shows that the friction between aluminum and itself is similar to the G10 fiberglass on it. The rails are fixed to the eyebolt bracket at the top and the deployer bulkhead at the bottom using 2 #4-40 screws in each location. These joints should only bear load from small deflections during load application to the eyebolt.

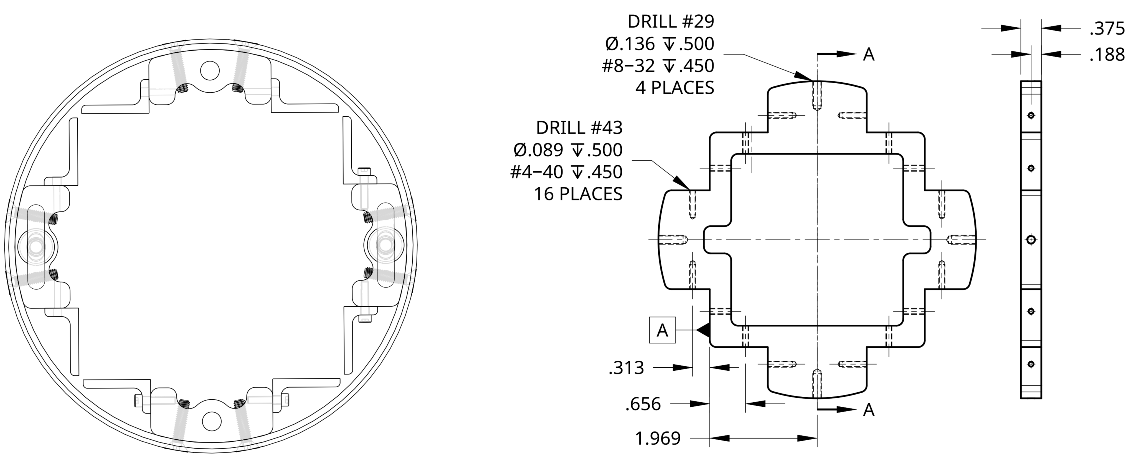
<u>Load Transfer</u>

Load will be applied to the deployer during three distinct events. Once during drogue separation, once during drogue snatch, and once during main snatch. During drogue separation, the forward CO_2 cartridges housed in the avionics bay will be triggered one at a time and release gas into the forward airframe. This will create pressure against the recovery unit and the walls of the rocket as the gas expands until the shear pins that keep the two sections together break. A small amount of force will be experienced at this point, given an excess of force to create a safety factor to break the shear pins. A tension force will act from the recovery unit to the bottom side of the eye-bolts on the deployer. Given the 500 lb rating of the eyebolt, the forces experienced at this point will be completely within safe loading conditions. From FEA, assuming a yield strength of 200×10^6 MPa, the eyebolt experienced a max stress of 136×10^6 MPa. This gives a factor of safety of 1.47.

During drogue deployment, we expect a larger tension force to be experienced for the recovery unit to the deployer eyebolts. This load will be applied in the designed direction for the eyebolt, which is rated for 500 lbs. With an expected snatch force of 100 lbs from drogue spread across two eyebolts, this connection has an estimated factor of safety of 2.



Left: eyebolt load application; Right: drogue loading FEA



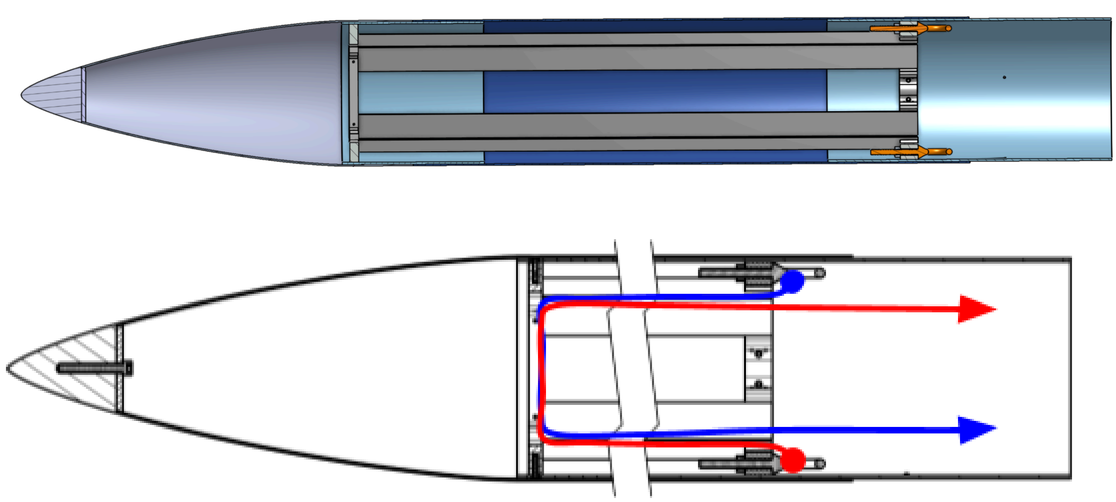
Left: Aft end of deployer wireframe — Screws shown through airframe.
Right: Deployer Bulkhead drawing

Cord Routing

To facilitate CubeSat deployment without the need for additional electronic sensing and deployment mechanisms, we are harnessing the forces from separation and drogue deployment to release the unit.

Two shock cords will run from a single eyebolt on the recovery unit, around The Log on opposite sides, terminating on opposing deployer eyebolts. As mentioned previously, the forces from drogue separation and deployment will put these lines in tension and force the CubeSat upwards along the rails.

After the lines straighten and The Log is deployed, the shock cords will connect the forward airframe to the recovery unit and hence the rest of the rocket during descent.



Top: Cross-section deployer and forward airframe
Bottom: Cord Routing with shock cords shown in red and blue (shortened)

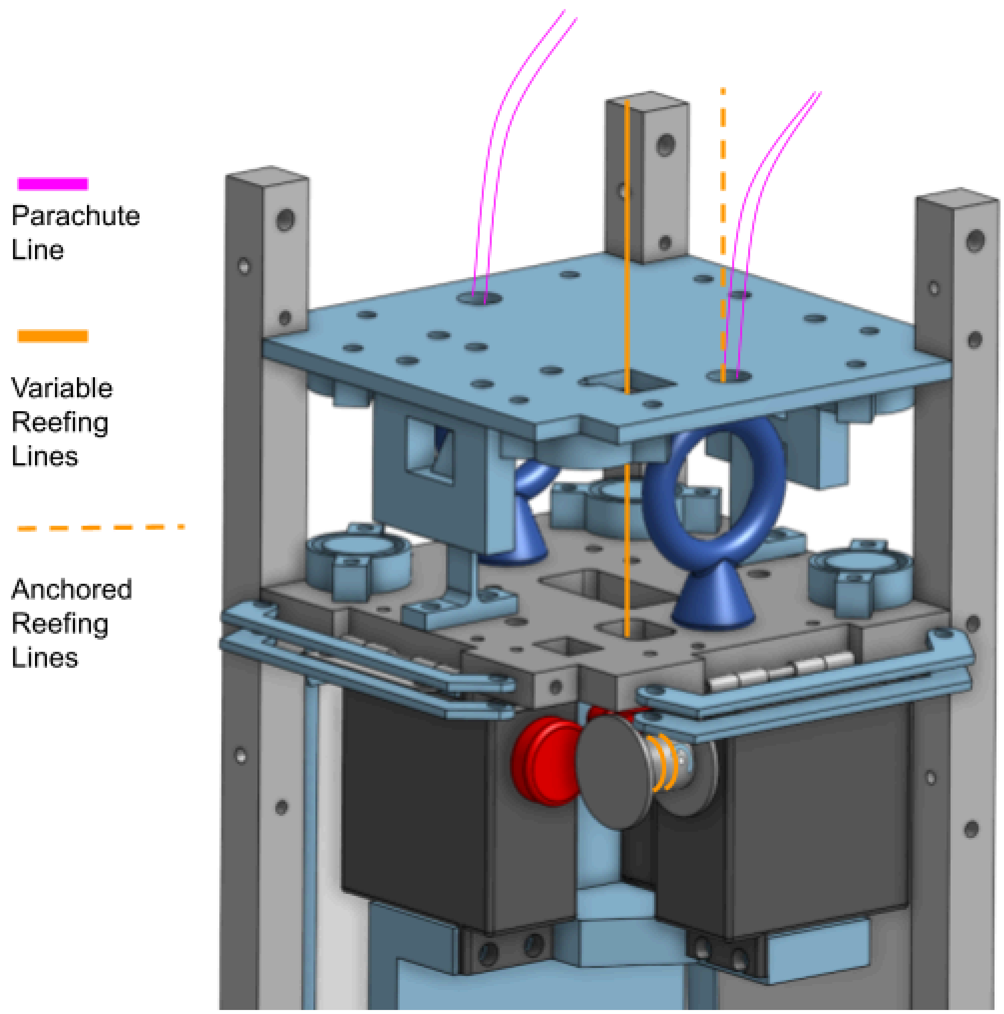
4. CubeSats

4.1. Helios

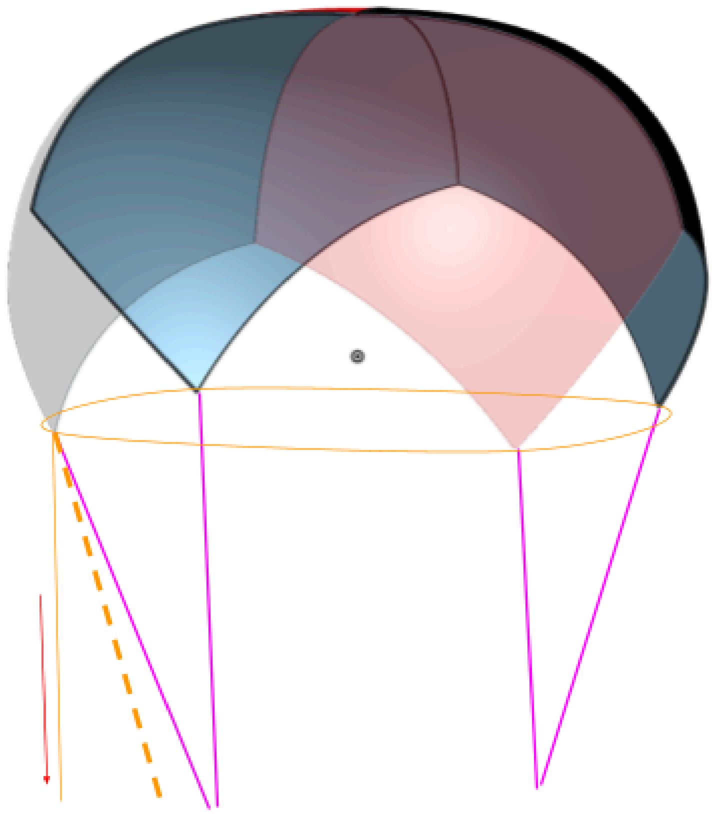
Variable Reefing Parachute System

Reefing is the act of constricting the aperture of a parachute in order to reduce the drag force and rock load produced by the parachute. Helios' Variable Reefing System (VRS) serves as a means by which descent speed can be controlled in order to account for changes in CubeSat mass.

A Rocketman 6 ft ultralight parabolic parachute will be deployed un-reefed as the main parachute for the “Log” , a 6U CubSsat composed of Helios and Hermes, immediately following separation at apogee. Following the separation event of Hermes from Helios, the VRS will reef the chute to account for the decrease in payload mass.



VRS: Payload Line Routing



VRS: Parachute Line Routing

The VRS system is a modified version of the common “skirt” reefing mechanism, in which lines are fed through the circumference of the parachute to statically restrict the diameter until the lines are cut. In the case of the VRS, the reefing ratio will be controlled through the rotation of a servo winch.

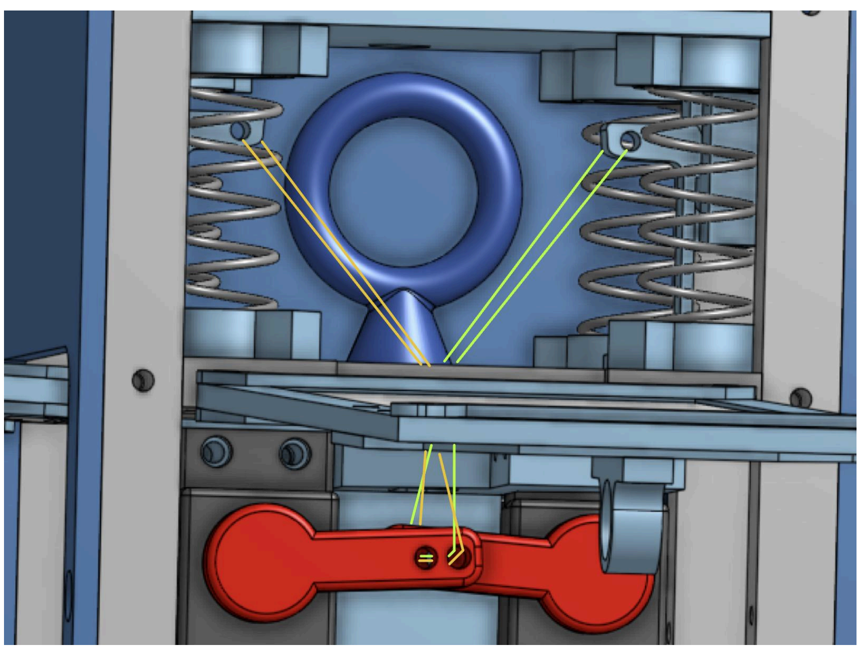
System testing was delayed due to the loss of the old parachute.

1. Parachute: Rocketman 6ft ultralight parabolic parachute
    a. Achieves target descent velocity at terminal speeds for 6U Log
    b. Four total lines; two will be anchored to each eyebolt symmetrically
        i. Eyebolts: 500 vertical load capacity → 2.11 Factor of safety through hand calculations. Chose higher factor of safety because Rocketman does not provide parachute dimensions outside of diameter
    c. Line (orange dashed) is anchored on the eyebolt, fed through additionally sewn cloth loops, and secured to the servo winch (orange)
    d. Servo winch (30 kg-cm) rotates counter-clockwise to close and clockwise to reef and disreef the chute

Release Mechanism Updates

Aluminum buckles were waterjetted to test stiffness. Using them as buckles in the manner outlined by the PDR mechanism was not successful. The desired application necessitated that the buckles deform beyond their permissible strain, causing failure. However, PLA buckles were tested earlier in the semester, and these worked for the given application. This test also revealed that the servos have enough pulling distance to actuate the buckles using the aluminum horns that they are bought with, so no additional horns will have to be machined. PLA buckles were tested about 15 times, and failure/fatigue did not occur over the course of these trials, indicating that thermoplastics can serve as a suitable material.

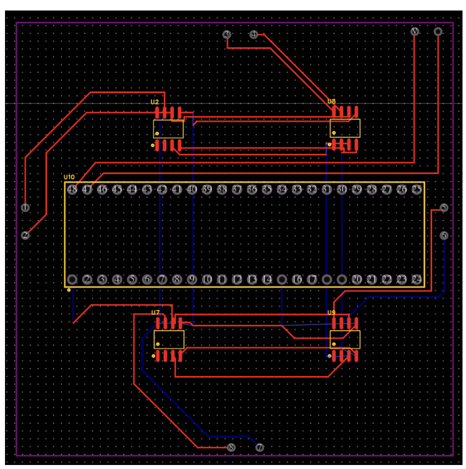
In order to reach a compromise between high resilience and low stiffness, 3D Printed PETG buckles will be employed. These will be printed on their side to avoid shear/tensile stresses along the direction of print layers. The mechanism for release will be the same design as presented in the PDR. The design is such that only 1 of the 2 servos needs to actuate to release both buckles. This design is active instead of passive to keep the CubeSat within the CubeSat standard, as any external retaining hardware in the rocket would violate the standard. It also allows for slight delay in parachute release after separation from the rocket, which helps to ensure that the CubeSat does not interact with the rocket upon release.



Line Routing for Buckle Actuation

Solar Panels/Experiment

1. Solar panels of varying chemistries (Perovskite, Monocrystalline silicon, polycrystalline silicon) will be used on different faces of the CubeSat
    a. Attempts to find GaAS solar panels (popular in space applications) have been generally unsuccessful due to its high expense
2. Open circuit voltage, short circuit current measured by custom PCB
    a. Results saved to SD card. Normalized efficiency calculated/analyzed after launch to determine efficiencies/degradation of different panel types over time after rocket launch and use in arid conditions (which can be framed as being representative of a Martian mission).
    b. The PCB will make use of an IA219, a high-side current shunt and power monitor IC (integrated circuit), to measure power produced by the solar panel.
        i. The PCB's location in the Helios has not been fully finalized however it will be horizontally placed somewhere in the top end of Helios
    c. A single integrated PCB will include four IA219s, one for each PCB which will also include a Teensy 4.1 to handle and store the data onto an SD card (See image below)



PCB Design for Solar Panel Power Readings

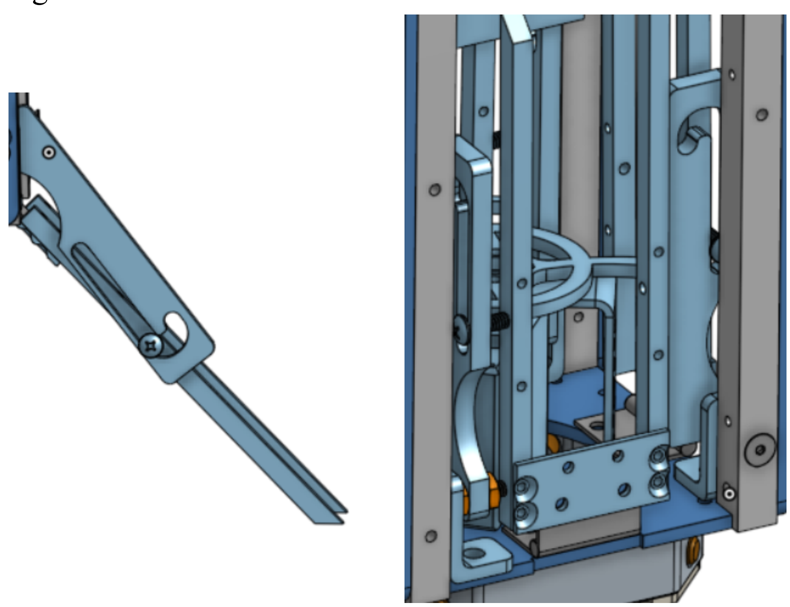
3. Code
    a. The IA219 will require C++ code to program (*See Appendix A for full code*)
    b. Summary: The code reads current, voltage, and power data from these sensors and logs it onto an SD card for analysis. In the setup phase, the INA219 sensors are configured, the SD card is initialized, and an organized data table is established on the SD card for storage.. In the loop, the current readings, along with voltage and power, are obtained for each sensor with a time, and formatted into the SD card.

This data string is then written to the SD card file. The loop continues with a delay between iterations

4. Waiting to hear back on use of Perovskite panels (Dr. Mitzi).
    a. Have been tested in space on the ISS, so their use is of high interest to the aerospace field. May become competitive with GaAs panels that dominate the market now.

Landing Legs

- Released in conjunction with the free rotating solar panel hatches via a central single servo actuator
- Sprung open via spring-loaded hinges
- Total of 4 independently locking and sprung legs
- Locking mechanism



Landing legs Unfolding, Folded

Bolt that keeps L-bracket for floor retention will also secure a slotted piece. A screw is set through this slotted piece and secured to a leg through a tapped hole. As legs extend, the screw pulls the slotted piece until fully extended. Contact with the ground moves the legs up and into a separate grooved section of the slot such that the legs lock in place.

Camera

Helios will be fitted with the 1080p 60fps Runcam Thumb used on last year's payload storing video locally to a removable SD card.

Power

Helios will be powered by one 2S LiPo battery. To power the computers, sensors, LoRa transmitter, and RunCam, a buck converter will step the voltage down to 5V. To power the reefing servo, the two chute release servos, and the other release servo (controls lumberjack, landing legs, and solar panels), another buck converter with a higher current rating will step the voltage down to 5V. The Feather S3 computer will measure the battery voltage using a voltage divider. This will be transmitted over LoRa.

Flight Computer

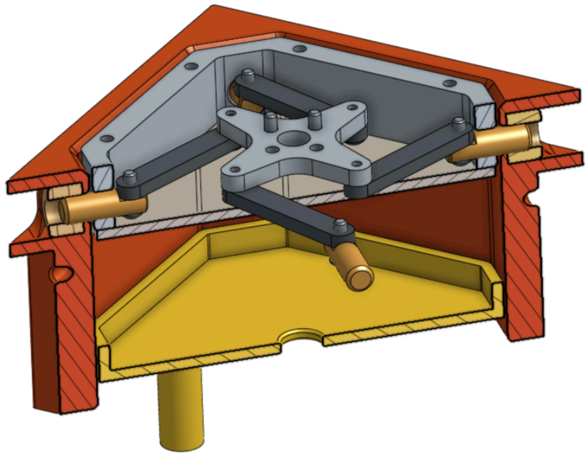
Helios shall include a flight computer to control parachute deployment, reefing, and telemetry. This will be controlled by a Feather-S3 microcontroller (in modem sleep mode to avoid unnecessary battery drain), and will include a PA1010D GPS module, LPS22HB altimeter, and BNO080 IMU for gathering data, as well as RFM95W LoRa transceiver for telemetry.

The gathered sensor data will be used to deploy the parachute a few seconds after apogee, trigger separation (Lumberjack), solar panels, and landing legs at 2500 feet, and vary reefing amount with altitude and speed. Parachute deployment will use two servos for redundancy, reefing will use a third, and separation/solar panels/landing legs will use a fourth. Telemetry will include all data from the sensors, the current battery voltage, and the status of parachute deployment, separation, and reefing, and will be sent over 915 MHz LoRa. Additionally, a COTS Featherweight GPS will provide redundant GPS telemetry.

The flight computer on Helios will also monitor battery voltage throughout the flight.

Lumberjack

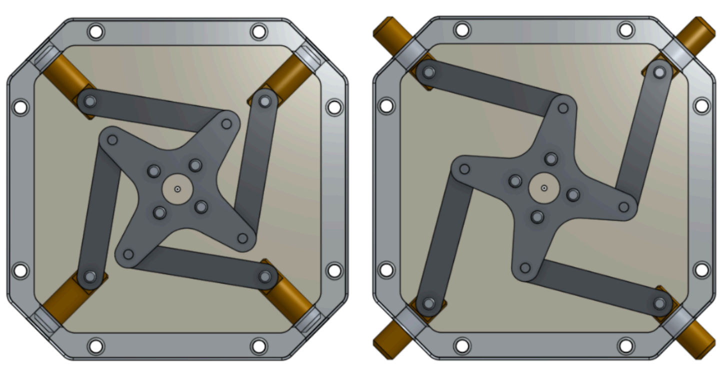
Cross-integrated between Hermes and Helios, the Lumberjack manages the separation event of The Log into two separate CubeSats. The separation system, hosted on Helios, is actuated jointly with the release of the landing legs and solar panels by the same 30 kg*cm dual-shaft output servo. The systems are synchronized by a 6061 aluminum sheet metal bracket joining the shafts.



Lumberjack cross-section

The Lumberjack adopts a pin-release separation system. Pins, 0.250” ± 0.005” in diameter and machined from 316 brass, are actuated by a 2-bar sliding linkage driven by the central servo. Each linkage is geometrically equivalent, with 4 total pins positioned at 45 degree angles from the Helios walls at each corner of the lumberjack, such that transmitted loads through the system are nearest the corner struts. The path of actuation of each pin is incident with the estimated center of mass for both CubeSats, mitigating rotational moments about the separation axis during the separation event.

The pins actuate through the machined 6061 aluminum guide that comprises the frame of the Helios component of the Lumberjack. Self-lubricating properties of 316 brass enable decreased system friction. The pins are received on Hermes by the Lumberjack receiver. Pins interact with bronze bushings press fit (externally) into the PETG receiver attached to the Hermes frame through #4-40 bolts fit on heat set threaded inserts to the walls and corner struts. The interior of the Hermes received serves as the actuation path of the recovery system spring plate which ejects the Hermes parachute once separation is complete. While stacked, the Hermes parachute will be pressed against the bottom of the lumberjack, upon which a polycarbonate plate is mounted to prevent snagging the parachute.



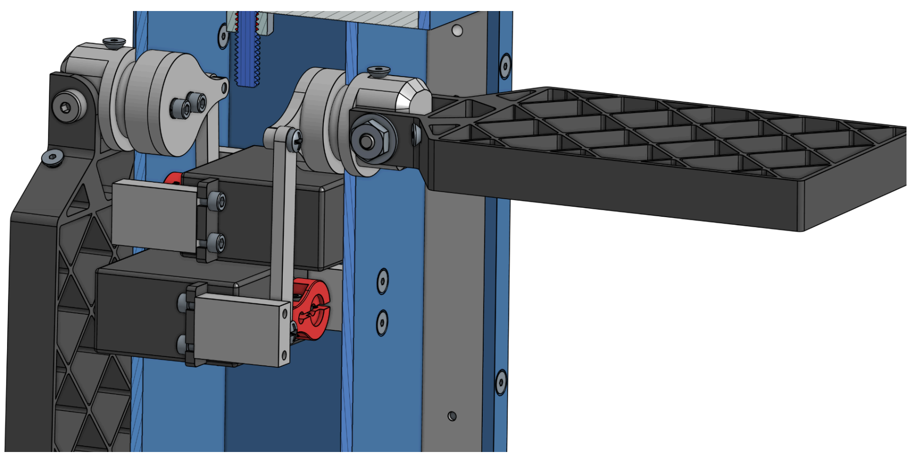
Lumberjack (Helios component) retracted (left) and deployed (right)

4.2. Hermes

Grid Fins

Hermes will have two grid fins on opposing walls, providing active roll stabilization while under parachute. They will be controlled by the Feather S3 implementing a control policy (either PID/EWMA) tuned using measurements from ANSYS fluent simulations. Once motion is stabilized, the efficacy of grid-fin control will be evaluated by inducing and eliminating roll, and by attempting to induce pitch.

For the mechanical system, the two servos will be attached to opposing walls in the CubeSat. A linkage will go from the servo horn to the grid fin "stub" which will provide a smooth pivot. From initial testing, we believe we will achieve a smooth bearing-like joint from the aluminum stub clamped around the fiberglass wall. However, if this is unsuccessful, we will use a brass bushing instead. The grid fins are initially retained in a folded position against the CubeSat walls by an angle bracket, and upon a slight rotation by the servo motor, they will move out from under the bracket and pop up using springs. A 3/16" shoulder bolt is used at this joint to provide a smooth actuating surface. All parts of the rotation system will be machined or waterjet out of 6061 aluminum for strength and rigidity.



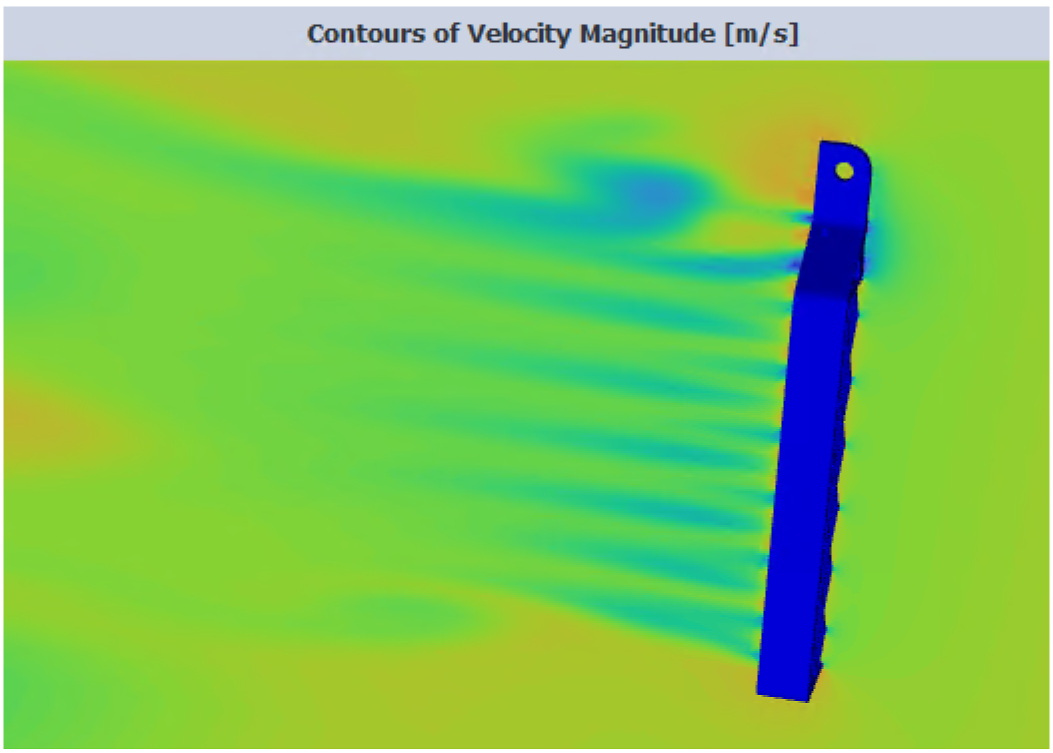
Hermes cross-section with full grid fin system shown

CFD studies have been run on this grid fin geometry. Of the studies run thus far, maximum force perpendicular to the flow direction occurs at the maximum expected inlet velocity (25 ft/s) and a deflection angle of 45°. The perpendicular force under these conditions is 0.2 N. While this does not sound like a lot intuitively, the rotational analog of newton's second law can be employed to reveal that this provides ample angular acceleration to the CubeSat. The calculation is shown below. A CFD result is also shown.

$$\sum_i \tau_i = 2rF = I\alpha \implies \frac{2 * 0.1384 * 0.2}{\frac{1.82}{12}(0.1^2 + 0.1^2)} = \alpha \implies \alpha = 517.12^\circ/s^2$$

Equation 1: Angular Acceleration in Degrees per Second Squared

The moment arm of the grid fin is the distance from the center of the grid fin to the centroid of the CubeSat body (found from CAD). The mass moment of inertia of a rectangle was employed using a mass of 4 lb, or equivalently 1.82 kg. The output is multiplied by a constant 2 factor to account for the two grid fins. This does neglect grid fin inertia, but that is likely so small as to be considered negligible due to their low mass.



Grid Fin under 25ft/s, 45-degree inlet velocity tilt

Live Video

Hermes will broadcast live video using a 5.8 GHz analog system. On the ground, a directional antenna and receiver will connect to a capture card, providing the video feed to a laptop over USB. In addition to live video, HD video from the same camera will be stored locally to a removable SD card on the payload.

Power

Due to dimension constraints, Hermes will be powered by two LiPo batteries. The computers, sensors, and the LoRa transmitter will be powered by a 1S LiPo regulated by the

Feather S3. The live video system and grid fin servos will be powered by a 3S LiPo, with a buck converter stepping to 5V for the servos. The Feather S3 will measure the voltage of both LiPos.

The range that we can get from our live video transmitter is severely reduced by heat buildup. To limit this, the live video components will be disconnected from the rest of the payload by a relay until deployment. This prevents heat build-up in the closed-off rocket, and means that the transmitter will be at its coolest when it is at its furthest from the receiver.

Flight Computer

Hermes shall include a flight computer to control grid fins and send telemetry data. This will be controlled by a Feather-S3 microcontroller (in modem sleep mode to avoid unnecessary battery drain), and will include a PA1010D GPS module, LPS22HB altimeter, and BNO080 IMU for gathering data, as well as RFM95W LoRa transceiver for telemetry.

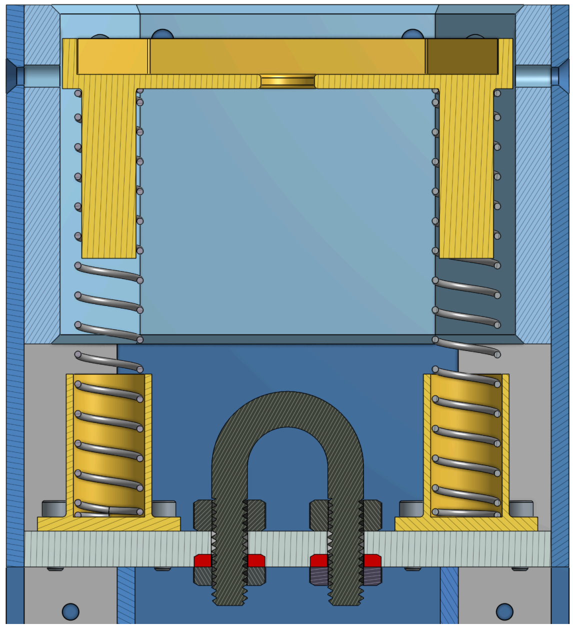
The gathered sensor data will be used to control the grid fins. First, roll will be stabilized based on the IMU gyroscope data. Then, changes in pitch and roll will alternately be induced, with results (time and degree of pitch and roll reached) being measured along with vertical descent rate and transmitted via LoRa. Additionally, telemetry will include all sensor data and current battery voltage sent over 915 MHz LoRa. A COTS Featherweight GPS will provide redundant GPS telemetry.

The flight computer on Hermes will also close a relay after initial payload deployment to power up the live video system. Battery voltages will be measured throughout and included in telemetry.

Recovery

The Hermes recovery system has been integrated into the 3U Hermes unit instead of as a separate 1U unit. Given the Lumberjack system, which automatically triggers Hermes parachute actuation from Helios, we deemed it unnecessary to attempt a 1U unit since it would no longer be standalone, instead relying on Helios for parachute deployment.

Upon Helios triggering Lumberjack actuation, the two units will separate and allow the spring plate under the parachute to extend and push the 4-ft parachute from Rocketman to deploy. This parachute is attached to an eyebolt screwed through a bulkhead inside Hermes by a Kevlar soft link which will be looped from the u-bolt to the parachute lines. This flexible connection allows the plate to compress to the U-bolt without the wasted space of a quick link on top. With a compressed plate, there is approximately 12 in^3 for the parachute to be folded into, which will be enough for the 4 ft parachute with an estimated folded volume of 9.8 in^3 to fit in.



Parachute spring plate system in yellow

5. CONOPS (Concept of Operations)

5.1. In-flight

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<u>The Log</u>

1. Rests in the deployer throughout launch up to the point of apogee
2. Once separation charges are fired by avionics, shock cords running from the recovery unit, under The Log, and back to eyebolts at the top of the deployer will become taut due to the forward airframe and avionics section moving apart.
3. Tautening lines will tighten around the bottom of The Log and propel it out of the deployer
4. The Log will free-fall for a few seconds (so it can move away from the rocket)
5. After detecting apogee, Helios' flight computers will wait a few seconds and then send a signal to two redundant servos to trigger VRS parachute release
6. Hermes's flight computers will trigger a relay, powering the grid fin servos and live video system.
7. The parachute will unfurl into its most disreefed configuration and continue falling
8. At 8,000 ft AGL, grid fin servos perform slight rotation to deploy grid fins
9. At 2,500 ft AGL the Helios flight computer will command the Lumberjack to actuate and split The Log into its Helios, Hermes configuration.

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<tr><td><u>Helios</u></td><td><u>Hermes</u></td></tr>
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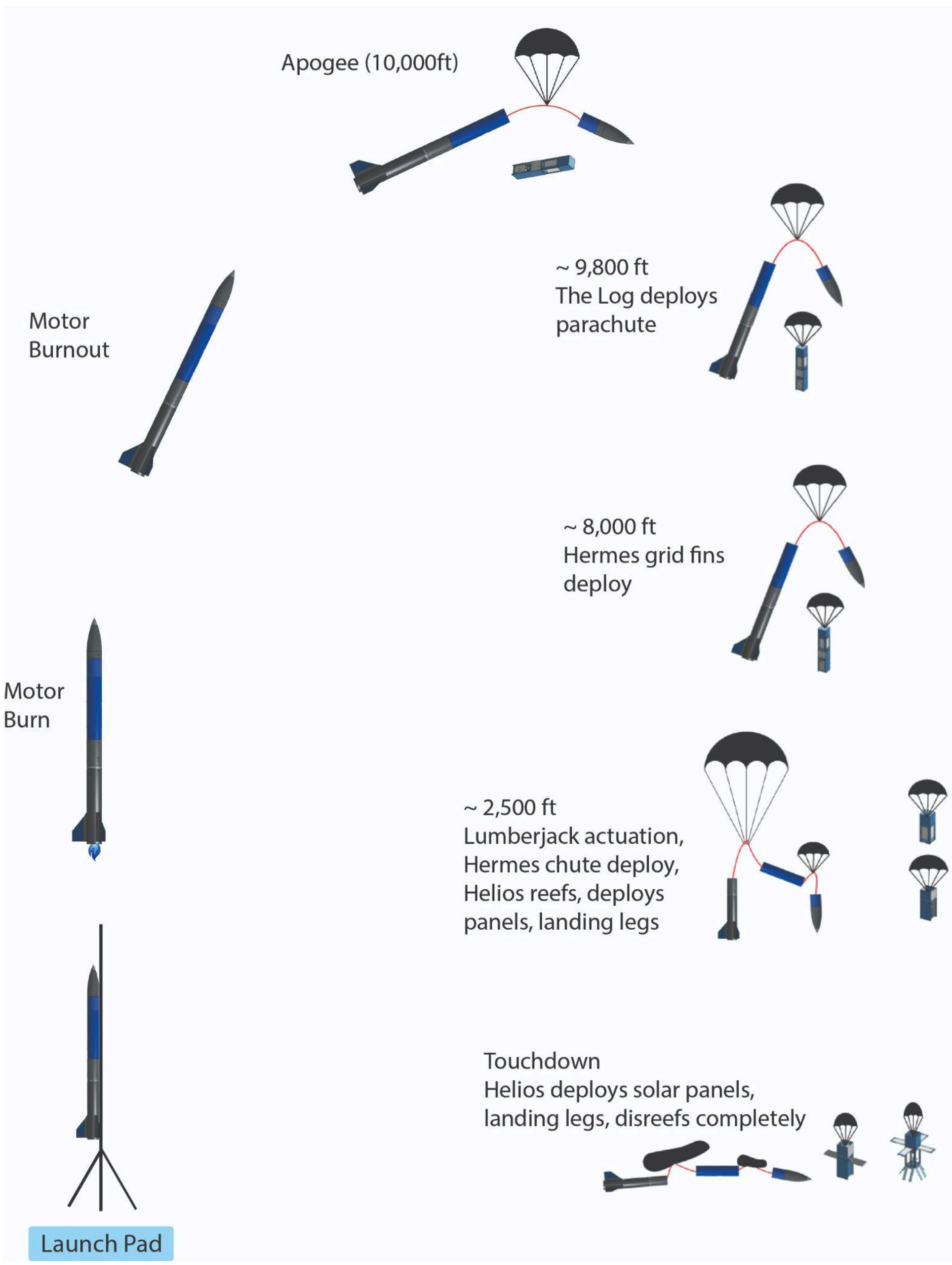
10. Lumberjack actuates: deploys solar panels, landing legs 11. VRS reefs to match Hermes descent velocity 12. Touchdown	8. Lumberjack actuates: passively deploys parachute 9. Grid fins begin roll stabilization control sequence 10. Grid fins alternate between inducing roll and pitch. 11. Touchdown



CONOPS Schematic

5.2. Post-Flight/Recovery

1. Disarm
2. Pull Data

6. Performance Metrics

The following metrics will indicate successful completion of each demonstration of technology/experiment.

Deployer

1. Cubesats are ejected in The Log configuration without affecting any rocket systems

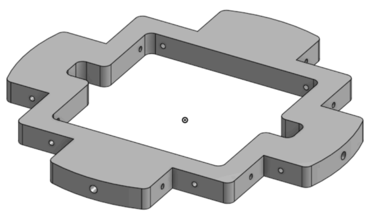
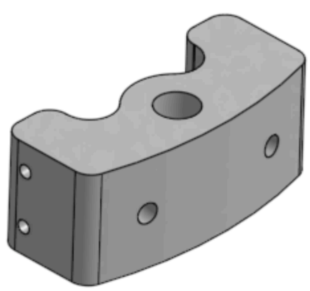
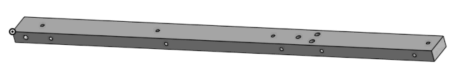
Helios

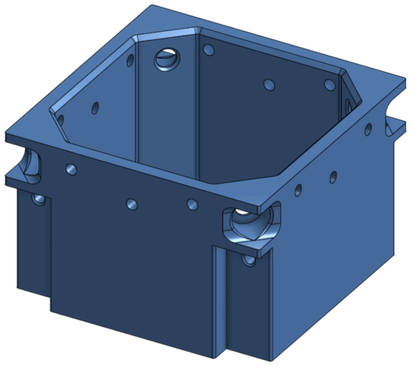
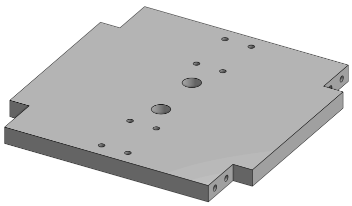
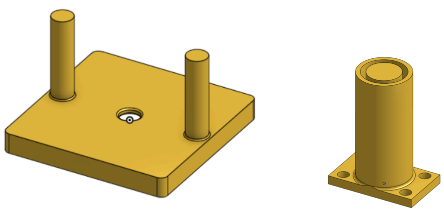
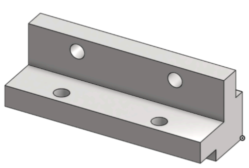
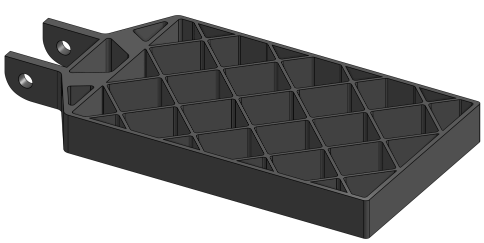
2. Variable Reefing System: Helios can increase its descent speed to within 10% of the expected value of Hermes' descent speed before descending below 1500 ft AGL.
3. Solar Panels: Deploy successfully. Data is logged successfully, efficiency comparisons can be made between the different types loaded onto Helios. Need efficiency data to be processed quickly enough to submit results within 4 hours of flight in worst case scenario for submission to judge panel.
4. Landing Legs: Deploy and lock successfully, do not break upon touchdown. Maintain angle within 10% of 45° upon landing.
5. GPS: GPS signal maintained through at least 80% of flight. Signal is steady upon landing.
6. Flight Computer: Executes reefing, parachute deployment, and GPS transmission successfully
7. Helios Camera: Logs data successfully to SD card for view after launch.
8. Power/Batteries: Able to maintain power supply over the entire duration of setup, flight, and recovery (10 hours).
9. Lumberjack: Splits the CubeSats successfully. No pins break.

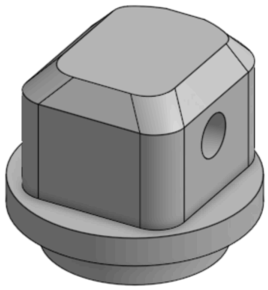
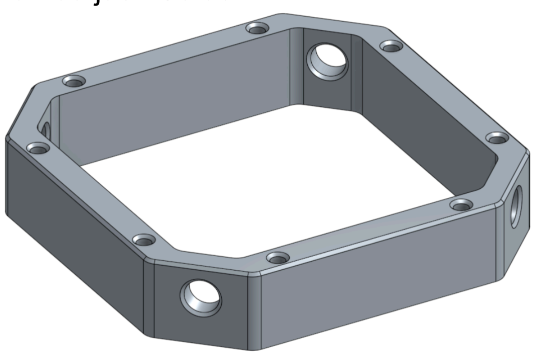
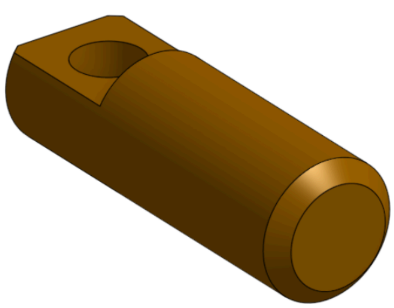
Hermes

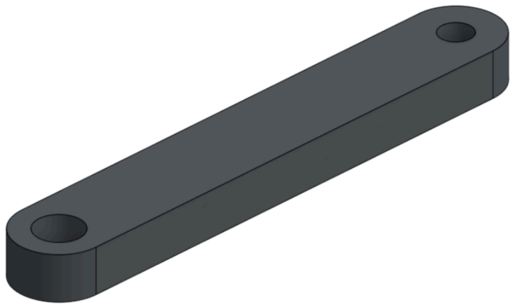
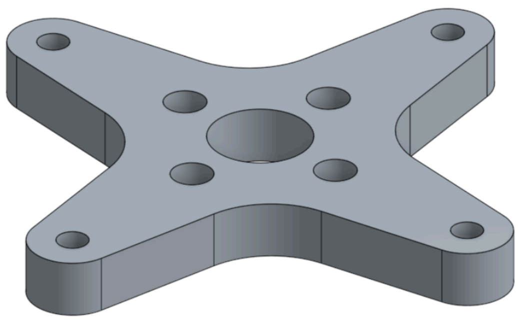
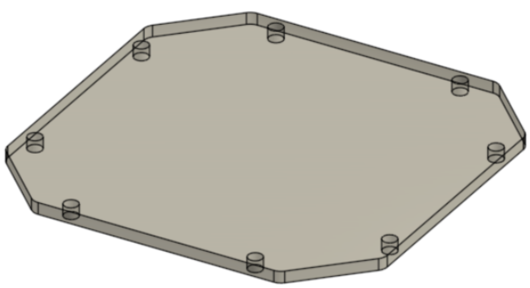
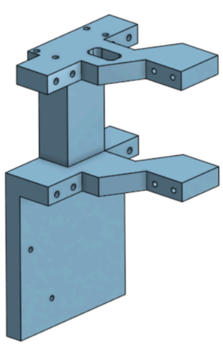
10. Live Video: Successful transmission of live video over at least 80% of flight duration
11. Grid Fins: Mitigate roll to no more than 10°/second.
12. Power/Batteries: Able to maintain power supply over the entire duration of setup, flight, and recovery (10 hours).
13. Flight Controller: Successfully executes roll/pitch control of grid fins.
14. Recovery: Parachute successfully opened and CubeSat falls with descent speed of no more than 15 ft/s.

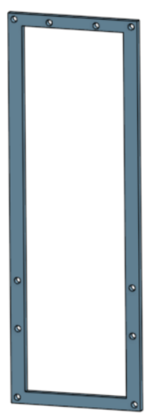
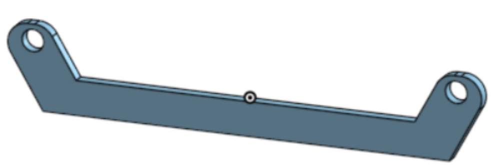
Parts/Manufacturing Plans

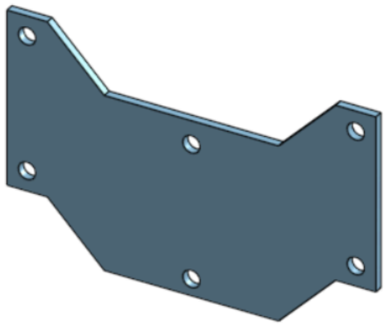
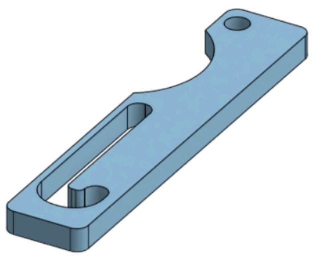
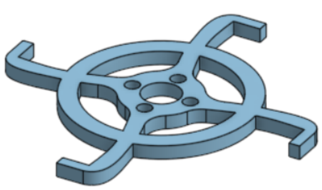
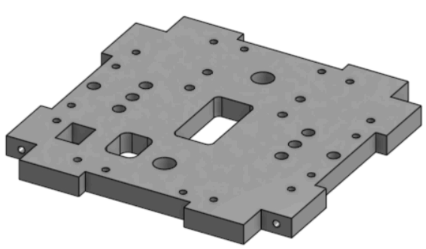
Deployer	
Part	Manufacturing Plan
Deployer Bulkhead (x1) 	Function: mounting point for deployer rails Material: 6061 AL ● Waterjet profile from .375” plate ● 3D print jig to drill holes in the mill ● Drill 5 holes on each side (4 sides) ● Tap 5 holes on each side
Eyebolt Rail Bracket 	Function: Mounting point for deployer eyebolt, mounting point for deployer rails Material: 6061 AL ● Mill profile ● Drill eyebolt holes ● Use 3d printed jig to drill body tube holes ● Drill rail holes
Hermes	
Corner Strut (x4) 	Function: Interior frame supports safe load transfer from bulkhead and walls Material: 6061 AL ● Cut 24” stock in half ● Mill 2 faces square ● Drill and tap holes on milled faces
Wall Frames (x4) 	Function: Encloses electronics and provides mounting points for small loading applications (camera, cord roller, etc.) Material: G10 Fiberglass ● Waterjet profile (including holes) ● Center drill countersinks
Lumberjack Receiver (x1)	Function: Smooth guiding surface for

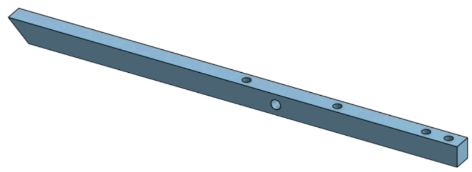
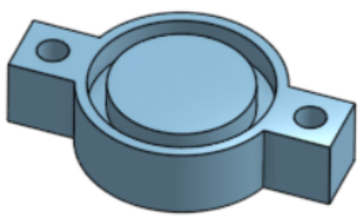
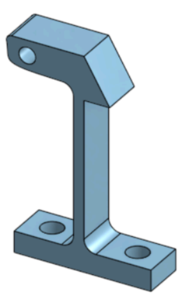
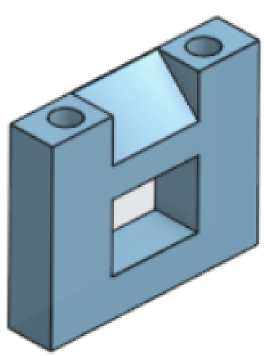
	parachute deployment, receives pins from Helios Lumberjack Material: PETG ● 3D Printed ● 4-40 Heatset inserts (x16) ● Brass bushings (x4)
Hermes Bulkhead (x1) 	Function: U-bolt attachment point, parachute spring plate mounting Material: 6061 AL ● Waterjet profile (excluding holes) ● Drill and tap small holes on face ● Drill U-bolt holes ● Drill and tap side holes
Hermes Chute Plate (x1) 	Function: Pushes parachute out of Hermes after Lumberjack actuation Material: PETG ● 3D print
Cord Roller (x2) 	Function: Provides smooth surface for deployer shock cords to slide along during ejection Material: PETG ● 3D Print ● Heat set inserts (x4)
Grid Fin (x2) 	Function: Provides active roll stabilization Material: Carbon Fiber Reinforced Nylon 6 ● 3D Print ● Heat set inserts (x4)
Grid Fin Stub (x2)	Function: Attachment point for grid fin,

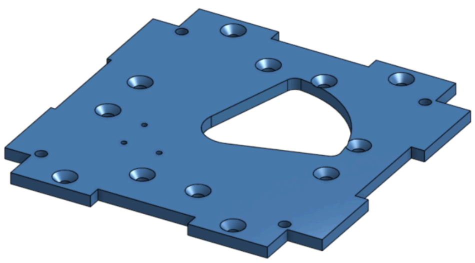
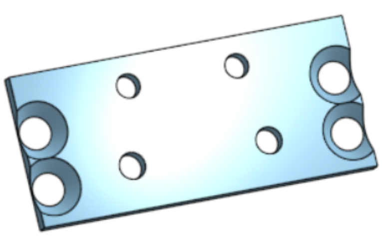
	provides smooth servo powered rotation Material: 6061 AL ● Stock: 1.15” round bar ● Face vise flat ● Clamp vise flat ● Mill top surface ● Flip in vise and mill bottom surface ● Drill 3/16” hole ● Drill and tap spring hole ● Drill and tap mounting hole
Helios	
Lumberjack Guide 	Function: Guides pins into Lumberjack reciever in Hermes Material: 6061 AL ● Milled from 0.75” stock (3.5”x3.5”) ● First Mounting (Primary profile) ○ Profile milled using 0.25” endmill ○ Clearance 4-40 thru holes ○ End chamfer using 45 degree chamfer end mill ● Second Mounting (Pin holes) ○ Waterjet machining jig to hold 45 degree angles ○ Reamed to 0.2500” ○ Minimal chamfer
Lumberjack Connection Pin 	Function: Locks Hermes and Helios together in Lumberjack system, smooth sliding fit Material: 316 Brass ● Turned to profile on lathe ● Holes and facing done in mill (mounted using 0.25” collet block)
Lumberjack Linkage	Function: Linkage between pins and servo horn adapter Material: 6061 AL ● Waterjet (except holes)

	● Drill clearance hole ● Drill and tap small hole
Lumberjack Horn Adapter 	Function: Mounting point for linkage and servo horn Material: 6061 AL ● Waterjet outline ● 3D print holding jig ● Drill and tap holes
Lumberjack Cap 	Function: Encloses lumberjack moving parts Material: 3/32” Acrylic ● Laser cut profile and holes
Electronics/Servo Mounting Plate 	Final Geometry TBD, dependent on rest of electronics Function: holds all electronics and the three top servos Material: PETG, 100% Infill ● 3D-Printed ● Size 4, Size 6 heat set inserts ○ Exact amounts TBD
Solar Panel Locking Piece (4x)	Function: Locks solar panels by fitting into servo actuation arms.

	Material: 6061 AL ● Waterjet cross section with large hole from ¼” Al stock ● 3D Print jig to hold upright ● Mount in drill press, drill the two through holes in jig
Solar Panel Bracing Piece 1 (4x) 	Function: Retains solar panels Material: ⅛” G-10 Garolite Fiberglass ● Waterjet profile and holes
Solar Panel Bracing Piece 2 (16x) 	Function: Retains solar panels Material: 1/16” 6061 AL ● Waterjet profile and holes
Solar Panel Bracing Piece 3 (4x)	Function: Retains solar panels Material: 1/16” AL ● Waterjet

	
Sliding Landing Leg Piece 	Function: Secures Landing Legs Material: 0.016” 6061 Al ● Waterjet Profile
Solar Panel Release Piece 	Function: Retains solar panels and landing legs as part of Lumberjack system Material: ⅛” 6061 Al ● Waterjet profile
Top Bracket 	Function: Take load from VRS system, mounting point for various systems Material: ¼” 6061 Aluminum ● Waterjet Profile, Holes ● Tap top face holes in vice ● Place sideways in mill ● Drill, Tap holes in side face in the mill ○ (has to be done 2x for the 2 holes on opposing faces)
Landing Leg	Function: Provides soft touchdown for Helios

	Material: 0.25” 6061 Al ● Waterjet profile with angled end with its one hole ● Tap the hole in vice ● Turn onto its top ● 3D print jig ● Mark, drill holes in drill press
Spring Retainers (8x) 	Function: Holds parachute spring in place Material: 3D Print (PETG)
Locking Buckles 	Function: Holds parachute plate down before deployment Material: 3D Printed with PETG (2x) ● Printed flat for correct layer orientation
Buckle Receiver (2x) 	Function: Holds parachute plate down before deployment Material: 6061 AL ● Waterjet cross section ● Drill, tap holes in mill ○ Do not need to be exact, can potentially be marked and drilled in the drill press ● Print jig to hold at 45 degrees ● Mill off slanted portion to create pressure angle for buckle

Fiberglass wall (4x) 	Function: Encloses parachute, provides smooth sliding surface along rails Material: ⅛” G-10 Garolite Fiberglass ● Waterjet cross section ● Countersink holes
Helios Containing Floor 	Function: Lumberjack servo mounting, landing leg mounting Material: ⅛” Garolite G-10 Fiberglass (1x) ● Waterjet Cross section ● Countersink holes
Landing leg adapting Piece 	Function: Attaches to the Spring-Loaded Hinge Material: 1/16” 6061 AL ● Waterjet ● Countersink holes

Corner Strut (4x) 	Function: Interior frame supports safe load transfer from bulkhead and walls Material: 6061 AL ● Cut stock in half ● Face Sides ● Drill, tap holes ● Rotate ● Drill, tap holes ○ All holes are through holes

7. Testing Plans

1. Vibration Testing: Use the shaker table available at Duke to vibrate the payload at a specific frequency to emulate flight vibration and test the payload's structural integrity
2. Separation Testing: Initially, we will perform a test on the ground to ensure that the actuator is working properly. If this is successful, due to the possibility of damaging manufactured components before testing the full assembly, we will next 3D print a pseudo Hermes to connect underneath Helios and test the structural integrity of Helios with a drop test. If this is successful, we will then proceed to performing a drop test of the entire assembly to ensure that the two payloads separate successfully.
3. Log deployment: Will collaborate with the recovery team for their testing to ensure that log deployment is successful and ensure that the recovery and payload systems are working cohesively
4. Parachute Deployment: Perform a drop test to determine if the parachute will deploy correctly upon communication from the servo
5. Landing Test: Perform a drop test to determine if the landing leg locking mechanism works correctly and stabilizes the payload once it lands

8. Integration and Assembly Plans

8.1. During the Spring

<u>Deployer</u>

1. The deployer will be assembled outside the forward airframe and then slid in after it is together. This starts by screwing the rails to the bulkhead using #4-40 screws at the base. These should be left loose so that the rails can move and allow the eyebolt bracket to fit in between them. This can be screwed on loosely as well. Then the eyebolt can be tightened with a ¼" nylon lock nut on the backside.
2. Use a square or flat surface (possibly put The Log inside to square rails) and tighten screws at both ends. Slide the deployer into the forward airframe and couplers. Put the

nosecone on and tighten screws through the nosecone and coupler into the bulkhead. Tighten the screws at the other end as well.

Hermes

1. Structural rigidity comes from the corner struts and the fiberglass walls. First, the grid fin assembly will be attached to the walls, which involves clamping the rotation piece to the large hole in the walls. Then the servos can be screwed in from the outside. Then two rails can be screwed onto each wall through the back using short countersunk #4-40 screws that go no more than halfway into the rail to leave room for a screw on the other side. The wall strips can also be attached at this point with screws through the other side of the rail.
2. Before attaching the two sides, electronics should be installed onto their mounting surfaces
3. The spring plate and spring guides can be installed on the bulkhead, as well as the U-bolt with nuts on each side.
4. Now, the wall and bulkhead can be attached to two rails on one side, which will give Hermes its form. These components must be attached at the same time due to the screws that run through both the wall and the bulkhead. Now, the cord rollers can be attached at the bottom, which will serve as a mounting point for the floor wall. The cord rollers and live video camera can be installed on the floor before it is screwed into the side walls.
5. The Lumberjack receiver can be slid into the top and secured with countersunk #4-40 screws through the corner struts and walls.

Helios

1. Assembly will work from the 2 main brackets (one on the floor, one supporting recovery about ⅔ of the way up the CubeSat out to the corner struts. Keeping the walls out of the assembly until the very end makes assembly much easier.
    a. For the floor bracket, the spring hinges will be assembled at all four sides with the landing leg assembly (no locking components yet). The lumberjack servo will also be mounted. L brackets will be installed as well. Once the lumberjack servo is mounted, all the actuation horns/lumberjack pieces can be installed onto the servo.
    b. For the recovery bracket, spring hinges for the solar panel release will be installed first on the bottom side.
    c. Next, buckles and Eye bolts will be installed on the top face. Lines will be run through the buckles to the underside of the plate.
    d. The main electronics mounting piece (with electronics, servos already on) will then be installed with screws running through the top of the plate. Lines from buckles to servos will be attached at this stage.

   e. Then, spring retainers and springs will be installed on the top face. The pressure plate will then be fastened to the other ends of the springs using more spring retainers. At this time, the servo winch line will be run through these two pieces.
2. With the subsystems connected to the two main brackets installed, the brackets themselves will be fastened onto the corner struts. This is trivial for the recovery bracket. However, when integrating the floor bracket, the locking mechanisms for the landing legs must be installed. To do so, place the bolt through the fiberglass countersink, corner strut, and L bracket. Then, secure the other end of the bolt with a nut. Now, place the locking piece onto the other side of the nut. Move the landing legs out of the way manually if need be. Then place a nut onto the other side of the locking piece to keep it somewhat secured in the bolt. Rotate the locking piece such that the hole is coaxial with the hole in the side of the landing leg. Fasten the panhead screw through the locking piece and into the tapped hole. The landing legs should now be fully installed. All other screws can no be installed into the struts through the wall countersinks. At this time, the solar panel assemblies will be fastened onto the spring hinges on the top recovery bracket.
3. Lastly, fold the landing legs and solar panels in (this may likely be a two-person job) and lock the servo to finish assembly.

8.2. Launch Day Pre-flight

Deployer

1. Tighten bulkhead retaining screws through airframe
2. Tighten eyebolt bracket screws through airframe
3. Tighten eyebolt nut
4. Wipe down rails to remove sand and debris
5. Attach shock cords to eyebolts after attachment to recovery unit

Hermes

1. Remove large wall
2. Validate that screw switch is in “off” position
3. Insert charged batteries
4. Check screw tightness for critical components
5. Replace large wall
6. Stow grid fins
7. Attach parachute to u-bolt using soft link and quick link
8. Fold parachute and secure in recovery section with temporary retaining rod over the top (until integration with Helios)

Helios

1. Remove one of four fiberglass walls, actuate the solar panel servo to open solar panels/landing legs (if not already open)
2. Validate that screw switch is in “off” position
3. Insert charged batteries
4. Check screw tightness for critical components
5. Make sure buckles for chute deployment are properly locked
6. Replace fiberglass wall, fold in/lock solar panels and legs
7. Fold and attach parachute using quick link
8. Retain parachute by installing breakaway lid

Integration

1. Retract Lumberjack
2. Integrate the Lumberjack by inserting Helios into Hermes (don’t pinch the parachute)
3. Initialize Helios momentarily to engage Lumberjack by turning the Helios screw switch
4. Turn the Helios screw switch to the “off” position
5. Slide The Log into the deployer such that the shock cords are crossed beneath it
6. Verify that the screw switches are lined up with access holes on the body tube

8.3. On Pad

1. Actuate screw switches
2. Live video will periodically transmit for 30 seconds, so that we can ensure the system is functional.

Appendix A: Solar Panel Power Reading Code for INA219s

```
#include <Wire.h>
#include <Adafruit_INA219.h>
#include <SD.h>

// Create instances of the INA219 class for each sensor
Adafruit_INA219 ina219_1;
Adafruit_INA219 ina219_2;
Adafruit_INA219 ina219_3;
Adafruit_INA219 ina219_4;

// Define the chip select pin for the SD card module
const int chipSelect = BUILTIN_SDCARD;

void setup() {
  // Start serial communication for debugging
  Serial.begin(9600);

  // Initialize the INA219 sensors with specific I2C addresses
  configureINA219(ina219_1, 0x41);  // Adjust the I2C address based on your setup
  configureINA219(ina219_2, 0x42);
  configureINA219(ina219_3, 0x43);
  configureINA219(ina219_4, 0x44);

  // Set the INA219 calibration values (may need to be adjusted based on your setup)
  ina219_1.setCalibration_32V_2A();
  ina219_2.setCalibration_32V_2A();
  ina219_3.setCalibration_32V_2A();
  ina219_4.setCalibration_32V_2A();

  // Initialize the SD card
  if (!SD.begin(chipSelect)) {
    Serial.println("SD card initialization failed. Check connections or replace the card.");
    while (1);
  }

  // Write header to the SD card
  writeHeaderToSDCard("sensor_data.txt",
"Timestamp,Current_1,Voltage_1,Power_1,Current_2,Voltage_2,Power_2,Current_3,Voltage_3,Power_3,Current_4,Voltage_4,Pow
er_4");
}

void loop() {
  // Read data from each INA219 sensor
  float current_1 = ina219_1.getCurrent_mA();
  float voltage_1 = ina219_1.getBusVoltage_V();
  float power_1 = ina219_1.getPower_mW();

  float current_2 = ina219_2.getCurrent_mA();
  float voltage_2 = ina219_2.getBusVoltage_V();
  float power_2 = ina219_2.getPower_mW();

  float current_3 = ina219_3.getCurrent_mA();
  float voltage_3 = ina219_3.getBusVoltage_V();
  float power_3 = ina219_3.getPower_mW();

  float current_4 = ina219_4.getCurrent_mA();
  float voltage_4 = ina219_4.getBusVoltage_V();
  float power_4 = ina219_4.getPower_mW();

  // Get the current timestamp (you may use a real-time clock or millis() for a simple example)
  unsigned long timestamp = millis();

  // Create a string with the sensor readings and timestamp
  String dataString = String(timestamp) + "," +
```

```
                      String(current_1) + "," + String(voltage_1) + "," + String(power_1) + "," +
                      String(current_2) + "," + String(voltage_2) + "," + String(power_2) + "," +
                      String(current_3) + "," + String(voltage_3) + "," + String(power_3) + "," +
                      String(current_4) + "," + String(voltage_4) + "," + String(power_4);

  // Open the SD card file and write data
  writeDataToSDCard("sensor_data.txt", dataString);

  // Add a delay before the next loop iteration
  delay(5000); // Adjust the delay based on your application requirements
}

// Function to configure INA219 with a specific I2C address
void configureINA219(Adafruit_INA219& ina, uint8_t i2cAddress) {
  // Set the INA219 address based on the provided I2C address
  ina.setI2CAddress(i2cAddress);

  // Initialize the INA219 sensor
  if (!ina.begin()) {
    Serial.println("Failed to find INA219 sensor, check wiring!");
    while (1);
  }
}

// Function to open the SD card file and write data
void writeDataToSDCard(const char* fileName, String data) {
  // Open the SD card file
  File dataFile = SD.open(fileName, FILE_WRITE);

  // If the file opens, write the data to the file
  if (dataFile) {
    dataFile.println(data);
    dataFile.close();
    Serial.println("Data written to SD card (" + String(fileName) + "): " + data);
  } else {
    Serial.println("Error opening file on SD card.");
  }
}

// Function to write a header to the SD card
void writeHeaderToSDCard(const char* fileName, const char* header) {
  // Open the SD card file
  File dataFile = SD.open(fileName, FILE_WRITE);

  // If the file opens, write the header to the file
  if (dataFile) {
    dataFile.println(header);
    dataFile.close();
    Serial.println("Header written to SD card (" + String(fileName) + "): " + header);
  } else {
    Serial.println("Error opening file on SD card.");
  }
}
```

Appendix B: Screw Calculation Code

```
import numpy as np
import math

alloy_sigma = 170000
alloy_tau = alloy_sigma * .6
stainless_sigma = 70000
stainless_tau = stainless_sigma * .6


screwDia = [.060, .073, .086, .099, .112, .125, .138, .164, .190, .216, .250, .375, .5]
screwArea = []
screwSize = ['#0', '#1-64', '#2-56', '#3-48', '#4-40', '#5-40', '#6-32', '#8-32', '#10-24', '#12-24', '1/4"-20', '3/8"-16', '1/2"-13']

for i in range(len(screwDia)):
    screwArea.append(np.pi * (screwDia[i]/2)**2)

maxLoad = float(input("Input max load (lb): "))
numScrews = float(input("Input number of screws: "))
material = input("Type 'a' for alloy steel and 's' for 18-8 stainless: ")

# maxLoad = 1000
# numScrews = 4
# material = 'a'

if (material == 's'):
    material_sigma = stainless_sigma
    material_tau = stainless_tau
else:
    material_sigma = alloy_sigma
    material_tau = alloy_tau


def ScrewCalc (numScrews, maxLoad, material_sigma, material_tau):

    nomLoad = float(maxLoad) / float(numScrews)
    A_sigma = float(nomLoad) / float(material_sigma)
    A_tau = float(nomLoad) / float(material_tau)

    for i in range(len(screwArea)):
        screwShear = nomLoad / screwArea[i]
        actualShear = nomLoad / A_tau
        fs = actualShear / screwShear
        if fs > 5:
            break

    minScrewSize = screwSize[i]



    print("\nMinimum screw size needed: " + minScrewSize)
    print("Factor of Safety: " + str(np.round(fs,3))+ "\n")



ScrewCalc(numScrews, maxLoad, material_sigma, material_tau)
```